

Xuan Luo

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Research Focus

I develop **adaptive and efficient language models** through:

- Adaptive computation in LLMs (**FlexiDepth**, **DiffSkip**, **TAR**).
- Efficient Attention Mechanisms (**KVPath**, **AHA**, **DAR**).
- Verification-free multi-token prediction (**DMTD**).

Education

University of California, Santa Barbara

Sep 2023 – ongoing

Ph.D. Student, Computer Science

Advisor: Prof. Xifeng Yan

Xi'an Jiaotong University

Aug 2019 – June 2023

B.E. Computer Science (Honors Program)

Manuscripts

KVPath: Connecting Key-Value Representations Across Transformer Layers

2026

Xuan Luo, Zhiyuan Wang, Hanzhong Cao, Xifeng Yan

Connects KV representations across layers, reducing KV cache by 37.5% relative to MLA while improving performance.

Learning When Not to Attend Globally

2026

Xuan Luo, Jiaming Shan, Wesley Truong, Kailai Zhang, Hanzhe Zhang, Xifeng Yan

[PDF]

Token-Adaptive Representation for Attention

2026

Zhiyuan Wang, Xuan Luo, Xifeng Yan

Selected Publications

Direct Multi-Token Decoding

2026

[COLM 2026] Xuan Luo, Weizhi Wang, Xifeng Yan

[PDF] [CODE] [MODEL] [WEBSITE]

Enables multi-token prediction without extra parameters, extra computation, or speculative decoding.

Dual Dimensionality for Local and Global Attention

2026

[NeurIPS 2026] Zhiyuan Wang, Xuan Luo, Sirui Zeng, Xifeng Yan

[PDF]

Local KV supports both memory and prediction, while distant KV's role is largely confined to memory, allowing reduced KV dimensionality with negligible performance loss.

Adaptive Layer-skipping in Pre-trained LLMs

2025

[COLM 2025] Xuan Luo, Weizhi Wang, Xifeng Yan

[PDF] [CODE] [MODEL] [DATASET] [WEBSITE]

Oral Presentation (top 24/418). Also invited to give a talk at NVIDIA Research

Enables dynamic layer skipping in pre-trained LLMs while retaining full benchmark performance, revealing how computational demands vary across token types.

DiffSkip: Differential Layer Skipping in Large Language Models

2025

[ACL Findings 2025] Xuan Luo, Weizhi Wang, Xifeng Yan

[PDF] [CODE]

Bot or Human? Detecting ChatGPT Imposters with A Single Question

2024

[COLM 2024] Hong Wang, Xuan Luo, Weizhi Wang, Xifeng Yan

[PDF] [CODE] [DATASET]

Experience & Projects

TikTok

Jun 2026 – Sep 2026

Research Scientist Intern

- Replaced standard attention with MLA in billion-scale recommendation LLMs; implemented and optimized the Triton kernel, reducing training memory usage by 42%.
- Stabilized the Muon optimizer to enable reliable convergence in training.

Professional Services

Reviewer: COLM 2026, COLM 2025, NeurIPS 2024, ACCV 2022